

Chapter 2

Periodic Table & Elements

Learning Objectives

- Define effective nuclear charge and ionisation energy of elements.
- Explain and predict the periodic variation in atomic size, ionic size, and ionization energy in periods and groups.
- Identify and classify the s-block, p-block and d-block elements.
 - Describe their chemical properties and chemical reactions.
 - Predict the products of some common reactions.
- Understand the mole concept and formula mass
- Relate gas volumes to mole concept
- Calculate the formula of compounds from combustion analysis or elemental analysis

Part 1

Periodic Table and Periodic Trends

Mendeleev's Periodic Table (1869)

- Mendeleev assumed that some elements had not been discovered.
- He left blanks in his table at atomic masses 44, 68, 72 and 100.
- **Noble gases were absent from Mendeleev's table.**
- William Ramsay discovered the noble gases between 1894 and 1898.

REIHEN	Gruppe I - R ² O	Gruppe II - RO	Gruppe III - R ² O ³	Gruppe IV RH ⁴ RO ²	Gruppe V RH ³ R ² O ⁵	Gruppe VI RH ² RO ³	Gruppe VII RH R ² O ⁷	Gruppe VIII - RO ⁴
1	H=1							
2	Li=7	Be=94	B=11	C=12	N=14	O=16	F=19	
3	Na=23	Mg=24	Al=273	Si=28	P=31	S=32	Cl=355	
4	K=39	Ca=40	-=44	Ti=48	V=51	Cr=52	Mn=55	Fe=56, Co=59 Ni=59 Cu=63
5	(Cu=63)	Zn=65	-=68	-=72	As=75	Se=78	Br=80	
6	Rb=85	Sr=87	?Yt=88	Zr=90	Nb=94	Mo=96	-=100	Ru=104, Rh=104 Pd=106, Ag=108
7	(Ag=108)	Cd=112	In=113	Sn=118	Sb=122	Te=125	J=127	
8	Cs=133	Ba=137	?Di=138	?Ce=140	-	-	-	- - - -
9	(-)	-	-	-	-	-	-	
10	-	-	?Er=178	?La=180	Ta=182	W=184	-	Os=195, Ir=197 Pt=198, Au=199
11	(Au=199)	Hg=200	Tl=204	Pb=207	Bi=208	-	-	
12	-	-	-	Th=231	-	U=240	-	- - - -

In 1913, Moseley analyzed the frequencies of X-rays emitted by the elements and discovered that the underlying foundation of the order of the elements was **atomic number**, not atomic mass.

Modern Periodic Table

PERIODIC TABLE OF ELEMENTS

Chemical Group Block

Periodic Table of Elements (Partial View):

Atomic Number	Symbol	Name	Chemical Group Block
1	H	Hydrogen	Nonmetal
2	He	Helium	Noble Gas
3	Li	Lithium	Alkali Metal
4	Be	Beryllium	Alkaline Earth Metal
11	Na	Sodium	Alkali Metal
12	Mg	Magnesium	Alkaline Earth Metal
17	Cl	Chlorine	Halogen
18	Ar	Argon	Noble Gas
19	K	Potassium	Alkali Metal
20	Ca	Calcium	Alkaline Earth Metal
21	Sc	Scandium	Transition Metal
22	Ti	Titanium	Transition Metal
23	V	Vanadium	Transition Metal
24	Cr	Chromium	Transition Metal
25	Mn	Manganese	Transition Metal
26	Fe	Iron	Transition Metal
27	Co	Cobalt	Transition Metal
28	Ni	Nickel	Transition Metal
29	Cu	Copper	Transition Metal
30	Zn	Zinc	Transition Metal
31	Ga	Gallium	Post-Transition Metal
32	Ge	Germanium	Metalloid
33	As	Arsenic	Metalloid
34	Se	Selenium	Nonmetal
35	Br	Bromine	Halogen
36	Kr	Krypton	Noble Gas
37	Rb	Rubidium	Alkali Metal
38	Sr	Strontium	Alkaline Earth Metal
39	Y	Yttrium	Transition Metal
40	Zr	Zirconium	Transition Metal
41	Nb	Niobium	Transition Metal
42	Mo	Molybdenum	Transition Metal
43	Tc	Technetium	Transition Metal
44	Ru	Ruthenium	Transition Metal
45	Rh	Rhodium	Transition Metal
46	Pd	Palladium	Transition Metal
47	Ag	Silver	Transition Metal
48	Cd	Cadmium	Transition Metal
49	In	Indium	Post-Transition Metal
50	Sn	Tin	Post-Transition Metal
51	Sb	Antimony	Metalloid
52	Te	Tellurium	Nonmetal
53	I	Iodine	Halogen
54	Xe	Xenon	Noble Gas
55	Cs	Cesium	Alkali Metal
56	Ba	Barium	Alkaline Earth Metal
57	La	Lanthanum	Lanthanide
58	Ce	Cerium	Lanthanide
59	Pr	Praseodymium	Lanthanide
60	Nd	Niodymium	Lanthanide
61	Pm	Promethium	Lanthanide
62	Sm	Samarium	Lanthanide
63	Eu	Europium	Lanthanide
64	Gd	Gadolinium	Lanthanide
65	Tb	Terbium	Lanthanide
66	Dy	Dysprosium	Lanthanide
67	Ho	Holmium	Lanthanide
68	Er	Erbium	Lanthanide
69	Tm	Thulium	Lanthanide
70	Yb	Ytterbium	Lanthanide
71	Lu	Lutetium	Lanthanide
72	Hf	Hafnium	Transition Metal
73	Ta	Tantalum	Transition Metal
74	W	Tungsten	Transition Metal
75	Re	Rhenium	Transition Metal
76	Os	Osmium	Transition Metal
77	Ir	Iridium	Transition Metal
78	Pt	Platinum	Transition Metal
79	Au	Gold	Transition Metal
80	Hg	Mercury	Transition Metal
81	Tl	Thallium	Post-Transition Metal
82	Pb	Lead	Post-Transition Metal
83	Bi	Bismuth	Post-Transition Metal
84	Po	Polonium	Post-Transition Metal
85	At	Astatine	Post-Transition Metal
86	Rn	Radon	Noble Gas
87	Fr	Francium	Alkali Metal
88	Ra	Radium	Alkaline Earth Metal
89	Ac	Actinium	Actinide
90	Th	Thorium	Actinide
91	Pa	Protactinium	Actinide
92	U	Uranium	Actinide
93	Np	Neptunium	Actinide
94	Pu	Plutonium	Actinide
95	Am	Americium	Actinide
96	Cm	Curium	Actinide
97	Bk	Berkelium	Actinide
98	Cf	Californium	Actinide
99	Es	Einsteinium	Actinide
100	Fm	Fermium	Actinide
101	Md	Mendelevium	Actinide
102	No	Nobelium	Actinide
103	Lr	Lawrencium	Actinide

Group
Period

Main group
Transition metal
(transition element)
Lanthanides
Actinides

Metal
Non-metal
Metalloid

- Alkali metal
- Alkaline earth metal
- Halogen
- Noble gas

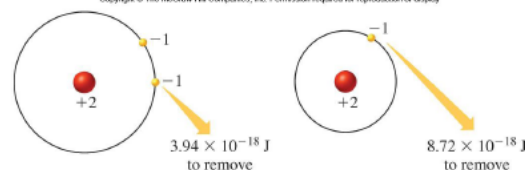
Periodic Table and Periodicity

- Certain properties of elements “repeat” in a periodic pattern.
 - For example, size of atoms decrease within a period but increases down a group
 - Ionization energy
 - Electronegativity
- Such pattern can be visualized using periodic table, together with the concept “effective nuclear charge”.

Effective Nuclear Charge (Z_{eff}) is the actual magnitude of positive charge that is experienced by an electron in the atom.

$$Z_{\text{eff}} = Z - \sigma$$

Z is the nuclear charge or simply the number of protons in the nucleus. σ is the shielding constant.



Z_{eff} increases from left to right across a period; increases very little down a group

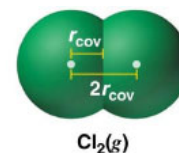
	Li	Be	B	C	N	O	F
Z	3	4	5	6	7	8	9
Z_{eff}	1.28	1.91	2.42	3.14	3.83	4.45	5.10

Periodic Trends in Atomic Radii

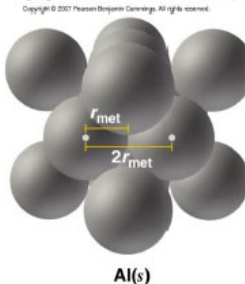
- The largest atoms are found in the lower-left corner of the periodic table and the smallest in the upper-right corner.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H 1																	He 2
2	Li 3	Be 4											B 5	C 6	N 7	O 8	F 9	Ne 10
3	Na 11	Mg 12											Al 13	Si 14	P 15	S 16	Cl 17	Ar 18
4	K 19	Ca 20	Sc 21	Ti 22	V 23	Cr 24	Mn 25	Fe 26	Co 27	Ni 28	Cu 29	Zn 30	Ga 31	Ge 32	As 33	Se 34	Br 35	Kr 36
5	Rb 37	Sr 38	Y 39	Zr 40	Nb 41	Mo 42	Tc 43	Ru 44	Rh 45	Pd 46	Ag 47	Cd 48	In 49	Sn 50	Sb 51	Te 52	I 53	Xe 54
6	Cs 55	Ba 56	La 57	Hf 72	Ta 73	W 74	Re 75	Os 76	Ir 77	Pt 78	Au 79	Hg 80	Tl 81	Pb 82	Bi 83	Po 84	At 85	Rn 86

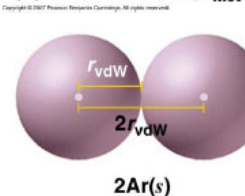
● Metals
● Semimetals
● Nonmetals



(a) Covalent radius, r_{cov}



(b) Metallic radius, r_{met}

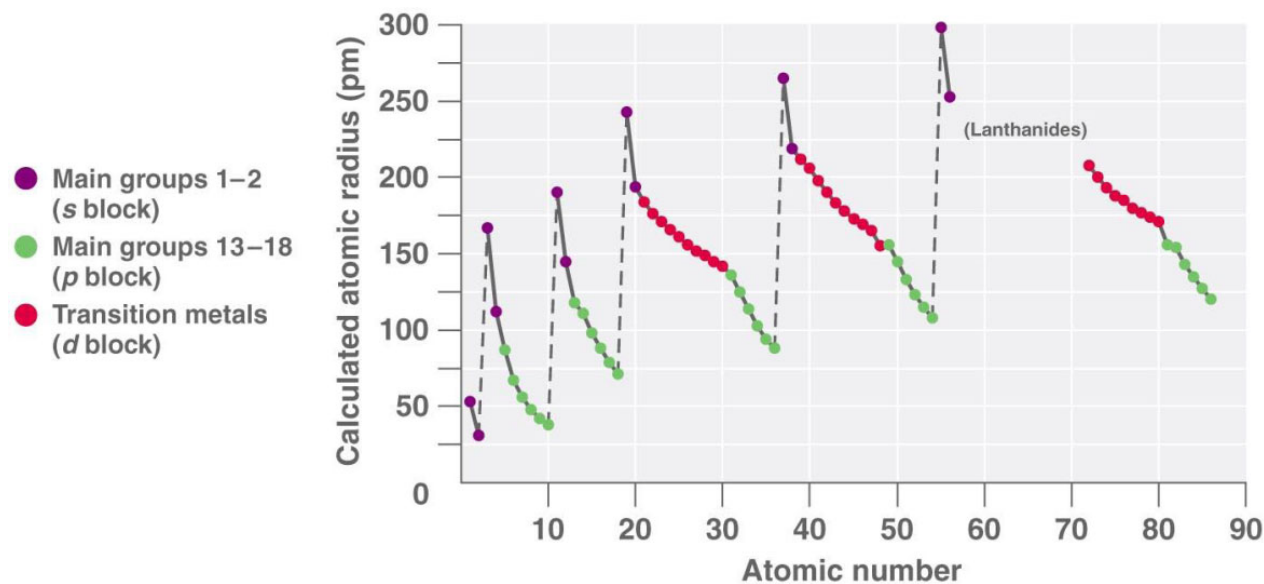


(c) van der Waals radius, r_{vdW}

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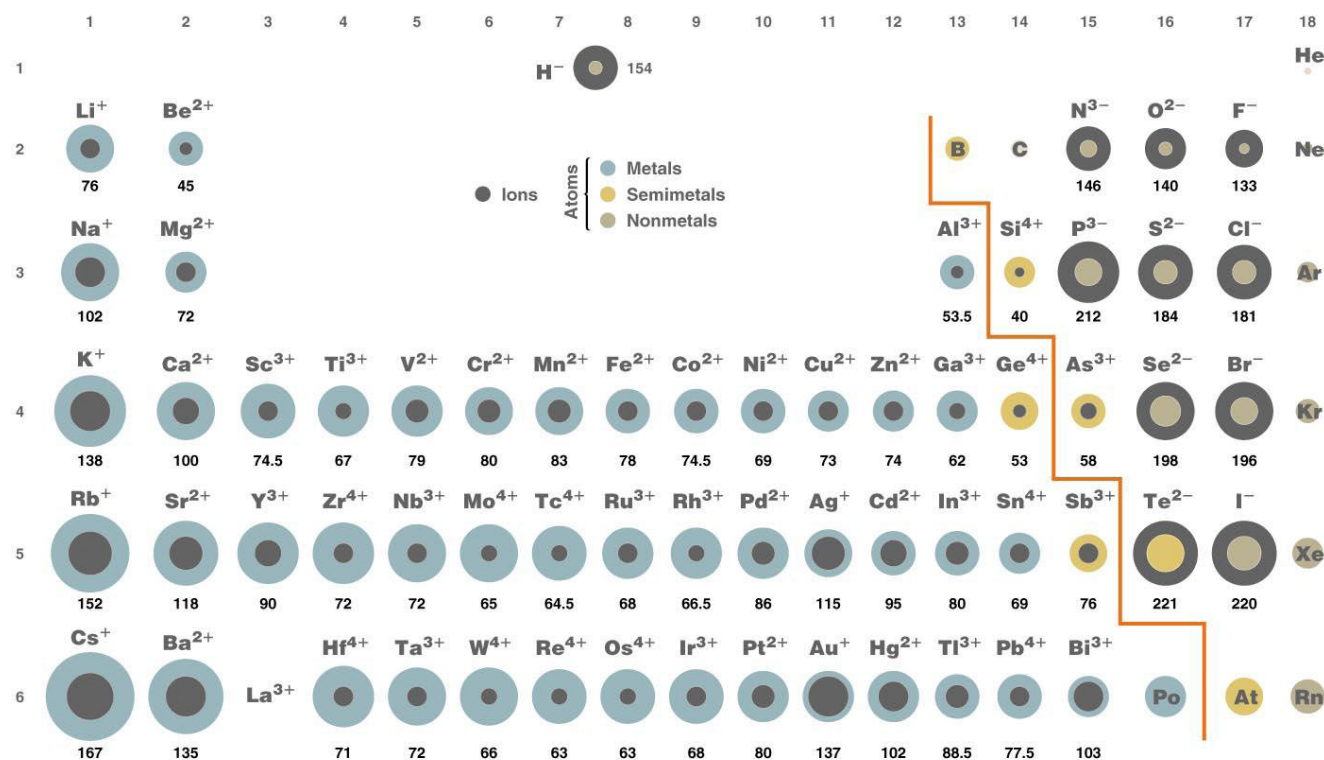
Periodic Trends in Atomic Radii

- The greater the Z_{eff} , the more strongly the outermost electrons are attracted to the nucleus and the smaller the atomic radius.
- Atomic radius is related to n^2/Z_{eff}
 - Moving across a period, n is constant, but Z_{eff} increases, so the radii decrease.
 - Moving down a group Z_{eff} increases, but n^2 increases more, so the radii increase.



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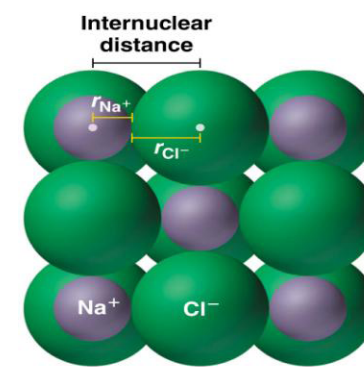
Ionic Radii and the Periodic Trend



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Cation vs atom

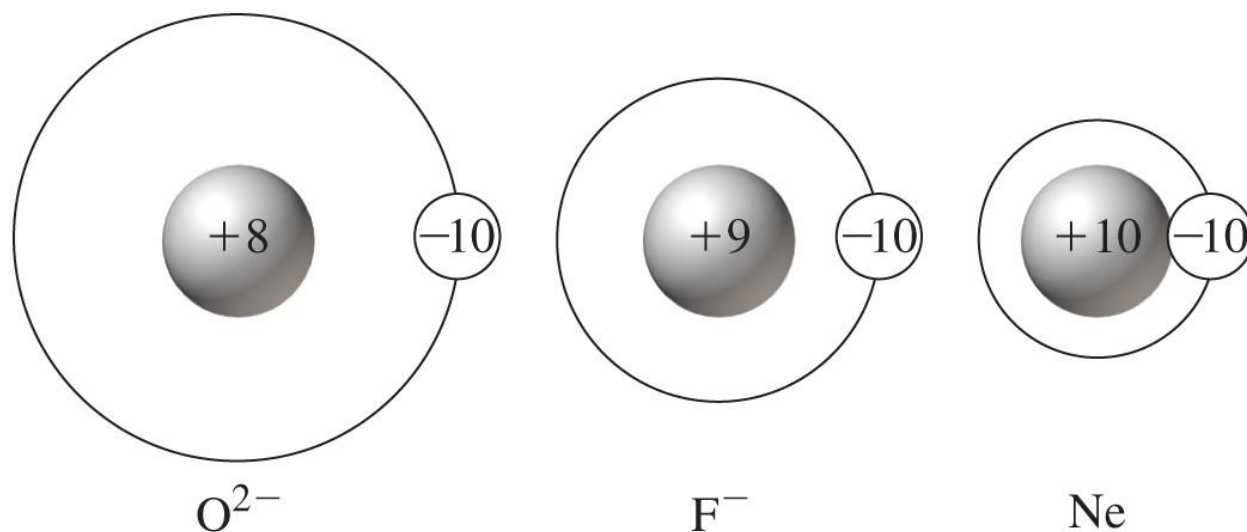
Anion vs atom



(a)
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Ionic Radii and Isoelectronic series

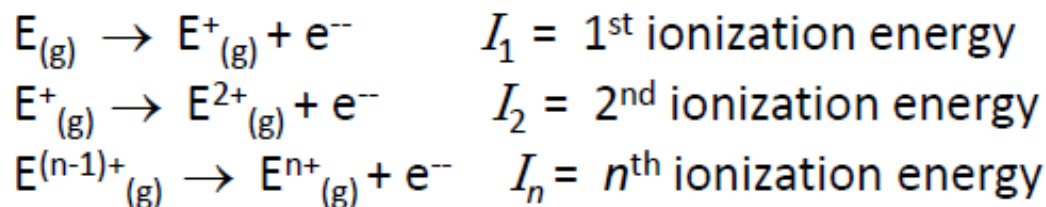
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Isoelectronic series is a series of two or more species that have identical electron configuration, but differ in their nuclear charges

Ionization Energy (IE)

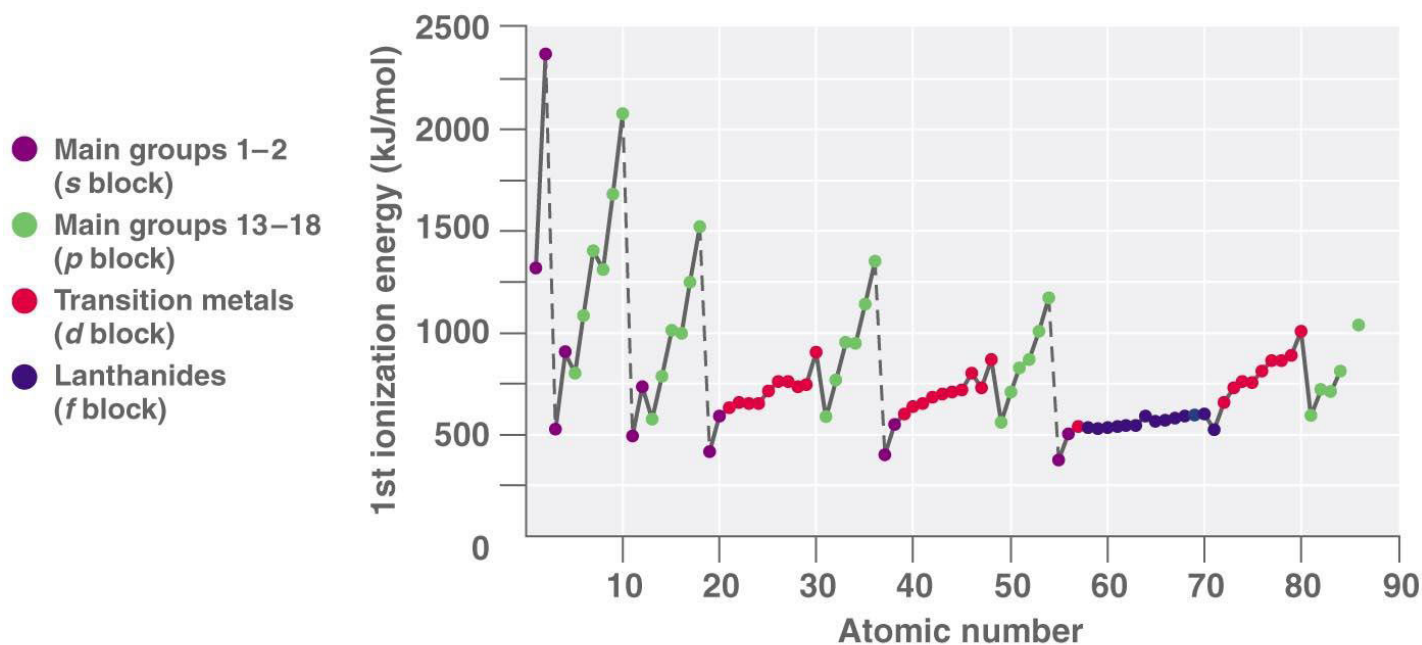
- Energy is required to remove an electron from an atom to form a cation.
- Ionization energy (I) is the amount of energy needed to remove an electron from the gaseous atom A in its ground state:
- $A(g) + I \rightarrow A^+(g) + e^-$ (energy required for reaction = I)
- Ionization energy is always positive ($I > 0$).
- Larger values of I mean that the electron is more tightly bound to the atom and is harder to remove.
- Units for ionization energies are kilojoules/mole (kJ/mol) or electron volts (eV): $1 \text{ eV} = 96.49 \text{ kJ/mol}$.



$I_2 > I_1$ because electron is attracted more strongly (electrostatic attraction) to E^+ than the neutral E atom.

In general, $I_n > I_{n-1} > \dots > I_2 > I_1$

Periodic trend of the 1st IE



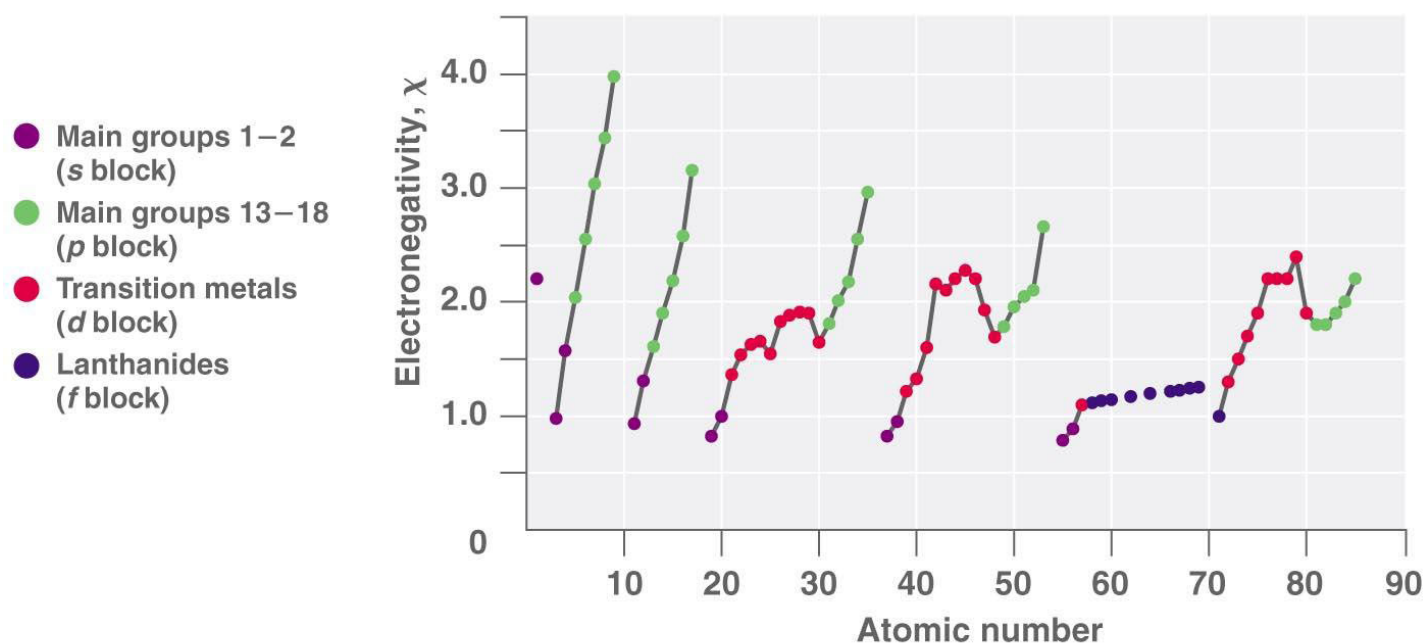
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1. In the same period (left to right), 1st IE generally increases. Z_{eff} increases, hence electrons are attracted more strongly.
2. In the same group (going downwards), 1st IE decreases. Valence electron is further from nucleus, hence attraction is weaker.
3. 1st IE of transition metals and lanthanides change slightly because their Z_{eff} is almost constant.
4. Higher IE (2nd, 3rd, ...) is more complicated, beyond the syllabus.

Electronegativity (EA)

- The ability of an atom to attract electron(s) towards itself in a chemical bond.
- Elements with high electronegativities tend to acquire electrons in chemical reactions and are found in the upper-right corner of the periodic table. Fluorine has the highest EA, 4.0.
- Elements with low electronegativities tend to lose electrons in chemical reactions and are found in the lower-left corner of the periodic table.

Periodic Trend of Electronegativity



- EA increases (from left to right) across a period
- EA decreases down a group

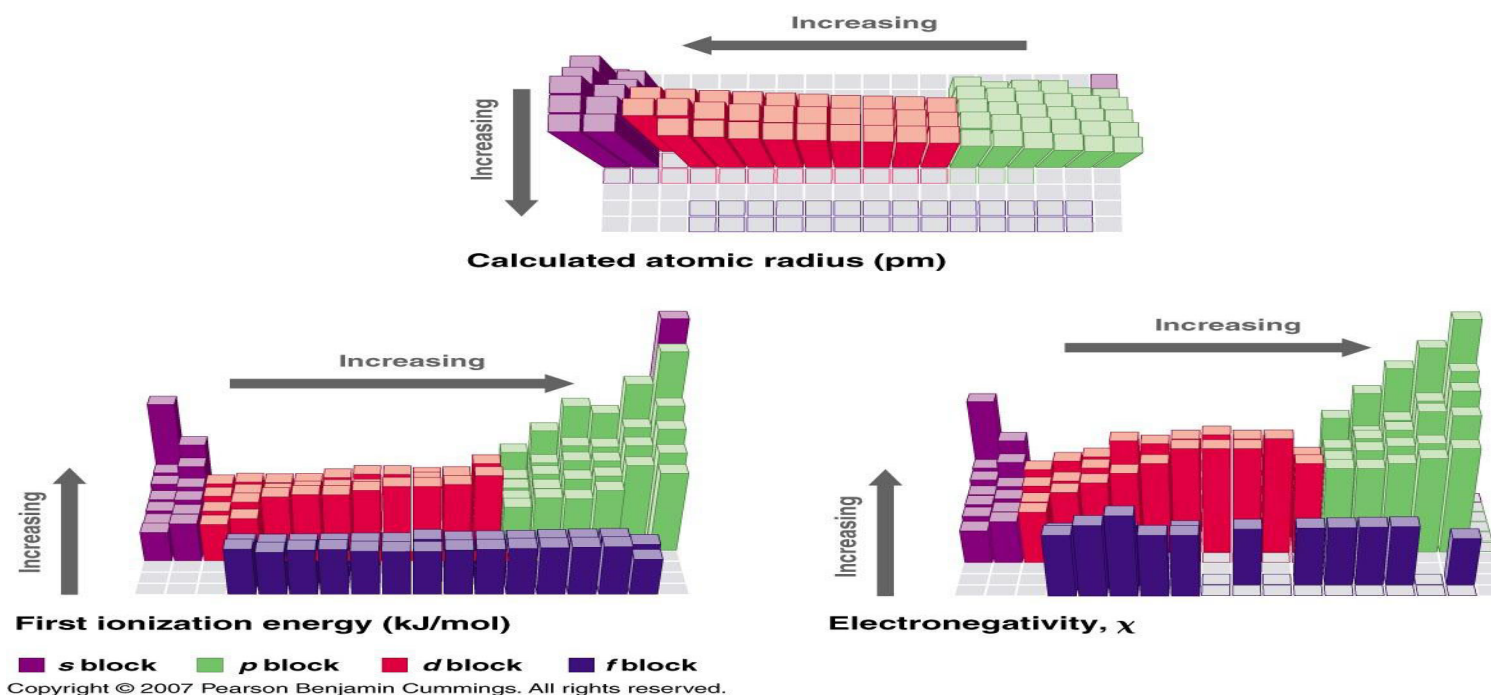
Electronegativity Chart

Using the Pauling scale

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period ↓																		
1	H 2.20																	He
2	Li 0.98	Be 1.57											B 2.04	C 2.55	N 3.04	O 3.44	F 3.98	Ne
3	Na 0.93	Mg 1.31											Al 1.61	Si 1.90	P 2.19	S 2.58	Cl 3.16	Ar
4	K 0.82	Ca 1.00	Sc 1.36	Ti 1.54	V 1.63	Cr 1.66	Mn 1.55	Fe 1.83	Co 1.88	Ni 1.91	Cu 1.90	Zn 1.65	Ga 1.81	Ge 2.01	As 2.18	Se 2.55	Br 2.96	Kr 3.00
5	Rb 0.82	Sr 0.95	Y 1.22	Zr 1.33	Nb 1.6	Mo 2.16	Tc 1.9	Ru 2.2	Rh 2.28	Pd 2.20	Ag 1.93	Cd 1.69	In 1.78	Sn 1.96	Sb 2.05	Te 2.1	I 2.66	Xe 2.60
6	Cs 0.79	Ba 0.89	* Lu 1.27	* Hf 1.3	* Ta 1.5	* W 2.36	* Re 1.9	* Os 2.2	* Ir 2.20	* Pt 2.28	* Au 2.54	* Hg 2.00	* Tl 1.62	* Pb 2.33	* Bi 2.02	* Po 2.0	* At 2.2	* Rn 2.2
7	Fr >0.79	Ra 0.9	* Lr 1.3	* Rf	* Db	* Sg	* Bh	* Hs	* Mt	* Ds	* Rg	* Cn	* Nh	* Fl	* Mc	* Lv	* Ts	* Og
			* La 1.1	* Ce 1.12	* Pr 1.13	* Nd 1.14	* Pm 1.13	* Sm 1.17	* Eu 1.2	* Gd 1.2	* Tb 1.1	* Dy 1.22	* Ho 1.23	* Er 1.24	* Tm 1.25	* Yb 1.1		
			* Ac 1.1	* Th 1.3	* Pa 1.5	* U 1.38	* Np 1.36	* Pu 1.28	* Am 1.13	* Cm 1.28	* Bk 1.3	* Cf 1.3	* Es 1.3	* Fm 1.3	* Md 1.3	* No 1.3		

Summary of Periodic Properties

1. Atomic radii decrease from lower left to upper right in the periodic table.
2. Ionization energy and electronegativity increase from the lower left to the upper right.



Part 2

Elements

Hydrogen

Hydrogen is the most abundant element in the universe: 74% by mass

On earth, hydrogen is only 0.9% by mass

- Grouped by itself
- Electronic configuration is $1s^1$

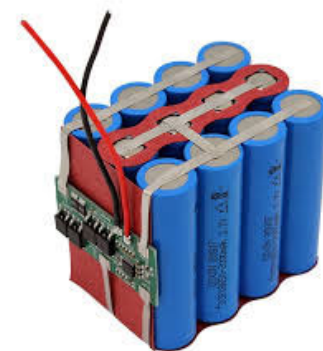
- Loses one electron to form a proton (H^+) – acid
- Accepts one electron to form hydride ion (H^-) – typically metal hydrides (MH_n)
- Metallic Hydrides react with water to produce hydrogen gas and a base.



- Form a covalent bond by sharing its electron with another atom:
 - BH_3 ; CH_4 , SiH_4 ; NH_3 , PH_3 , AsH_3 ; H_2O , H_2S , H_2Se ; HF , HCl , HBr , HI
- There are three isotopes: protium, deuterium, and tritium which differ in their physical properties such as melting point, boiling point, densities, heat of fusion and vaporization

Group 1 elements (alkaline metal)

- Reaction with water
- Reaction with oxygen
- Reaction with other non-metals
- Production of Group I metals
- Production of NaOH



Generally, soft and low melting point.
Also low ionization energy.

Na and K are abundant in nature,
others are less abundant. None of
them is in their atomic form in nature.

Group 2 elements (alkaline earth metal)

- Reactivity compared with Group 1
- Reaction with water and acid
- Reaction with oxygen (BaO_2)
- Reaction with nitrogen and other group-15 elements (M_3N_2)
- Common Group 2 compounds: MgO , CaCl_2 , CaCO_3 , CaO , Ca(OH)_2 , BaSO_4 , ...

Group 13 elements

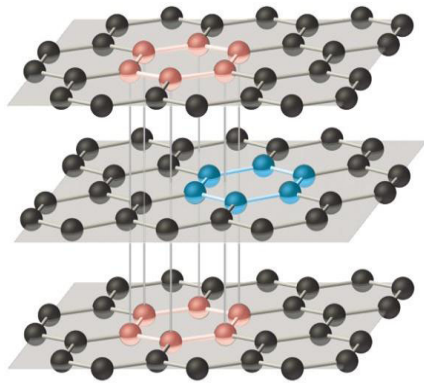
- BF_3 (and other boron halide): common Lewis acid (as catalyst)
- AlCl_3 (and other aluminium halide): common Lewis acid (as catalyst)
- Al_2O_3
- $\text{Al}(\text{OH})_3$
- NaBH_4 and LiAlH_4
- Production of Al:

Group 14 elements

- Carbon
 - diamond, graphite, fullerene, graphene (these are called _____)
 - CO, CO₂, metal carbonates
 - Metal carbide: CaC₂
 - Organic compounds: alkane, alkene, alkyne, alcohol, carboxylic acid, ...
- Si (semiconductor): Si / SiO₂ / glass / cement / Si-containing minerals
- Pb (+2, +4): lead-acid battery

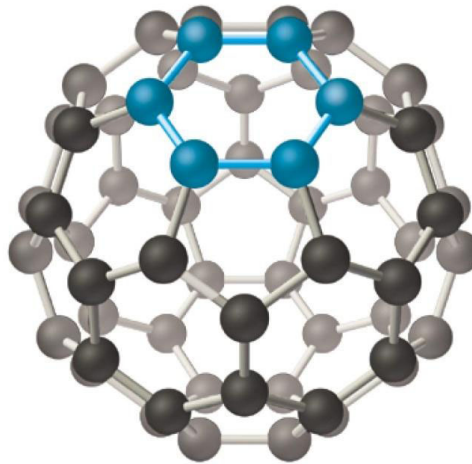


Structures of Carbon Allotropes



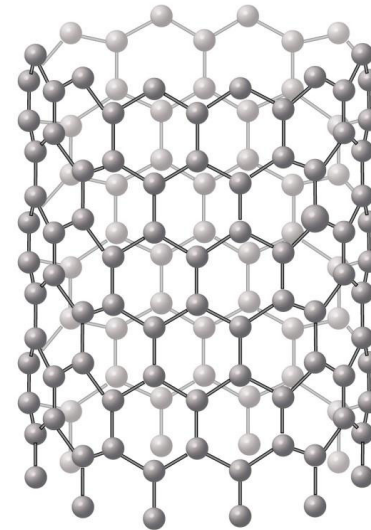
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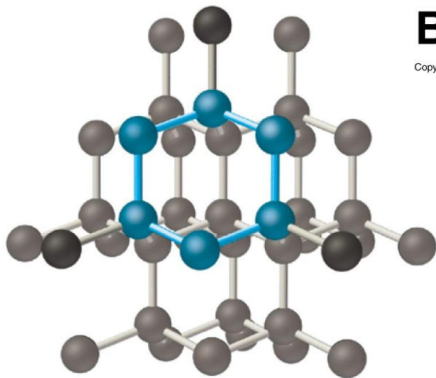


Buckminsterfullerene, C₆₀

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Nanotubes have carbon atoms in continuous hexagon arrangements rolled into tube like structures that form tiny fibers 10 to 12 times stronger than steel



Diamond

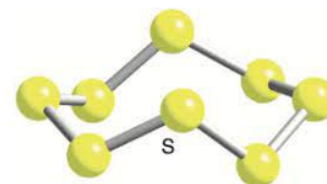
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Group 15 elements

- Nitrogen
 - Oxides (air pollution)
 - Ammonia and ammonium
 - Amino acid
- Phosphorus
 - Red / white phosphorus
 - Oxides
 - H_3PO_4 and salts
 - Agriculture use
 - DNA

Group 16 elements

- Oxygen
 - Allotropes of oxygen
 - Oxides: acidic / basic / amphoteric
 - Peroxide: H_2O_2 and _____
- Sulfur (S_8)
 - Oxides
 - Acids and salts
 - Hydride
 - Air pollution



Group 17 elements (halogen)

- Possible oxidation states
- Acid/salt without oxygen
 - HF
- Acid/salt with oxygen
- Substitution of halogens
- Identification of halide ions
- Production of Cl_2

Common transition metals

- Cr
- Mn
- Fe
 - Production
- Cu
 - Production
- Zn
- Ag

Preparation of common gases

- H_2
- CO_2
- N_2
- NH_3
- NO_2
- O_2
- SO_2

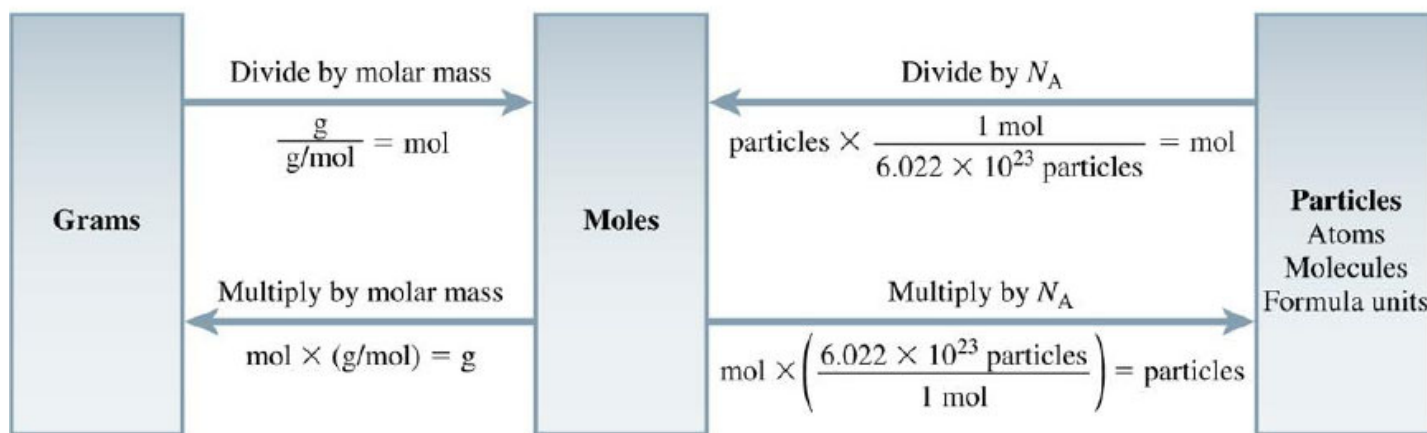
- H_2S
- Cl_2
- HCl
- CH_4
- C_2H_4
- C_2H_2

Part 3

Stoichiometry

Mole, molar mass and stoichiometry

- Avogadro's number
- Molar mass: atomic mass, molecular mass, formula mass



Empirical and Molecular Formula

- **Empirical formula** gives only the *relative numbers of atoms in a substance in the smallest possible ratio*.
- **Molecular formula** gives the *actual number of atoms of each kind present per molecule*.
- Percentage composition of the elements will be the same for empirical and molecular formula.
- We can determine the empirical formula of a compound using the percent composition. With other information, molecular formula can be deduced.

Combustion Analysis

- In the combustion of 18.8 g of a ternary organic compound **X**, 27.6 g of CO_2 and 11.3 g of H_2O are produced. Determine the empirical formula of **X** by calculation.

- Given that the molar mass of **X** is between 160 and 200, determine its molecular formula.

Elemental Analysis

- An unknown organic compound **Y** contains 53.9% C, 4.8% H, 7.9% N, 9.0% S, 6.5% Na and 17.9% O by mass. Determine its empirical formula.

Molar volume of gas

- 22.4 L/mol for any gas, at s.t.p. (standard temperature and pressure, 0 °C, 1 atm)
- But the value is different at different conditions:
 - 22.7 L/mol at 0 °C and 1 bar
 - 24 L/mol at 20 °C and 1 atm
- The safest way is simply use ideal gas law: $pV = nRT$
 - $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
 - Unit: $p - \text{Pa}$; $V - \text{m}^3$; $T - \text{K}$
 - Pressure: 1 bar = 10^5 Pa ; 1 atm = 101325 Pa

Balancing chemical equation

- Full equation vs ionic equation
 - KMnO_4 solution is unstable, so its concentration need to be determined by titration with potassium oxalate ($\text{K}_2\text{C}_2\text{O}_4$) in dilute sulfuric acid. Carbon element is oxidized to +4 oxidation state.
 - Write the balanced ionic equation of this titration.
-
- In this titration, hydrochloric acid should not be used to replace sulfuric acid. Write the balanced ionic equation to explain the reason.